Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP360B003-V1

Revision Record 修改记录

Rev No.版本号	Date 时间	Description内容
V.0	2022-05-10	First Release 首次发行

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Revision Record

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1.General Specifications 基本规格

No.	Item 项目	Specification 规格	Unit 单位	Remark
1	LCD Size 液晶面板尺寸	3.6”	inch	-
2	PanelType 面板类型	IPS	-	-
3	Resolution 分辨率	544xRGBx506	Pixel	-
4	Display Mode 显示模式	Normally Black	-	-
5	NumberofColors 颜色数量	16.7M	-	-
6	Viewing Direction 使用视角	ALL	-	Note1
7	NTSC 色彩饱和度	60%	-	Typ.
8	ContrastRatio 对比度	1200	-	
9	Luminance 亮度	1000	cd/m2	Typ.
10	Module Size 模组尺寸	98.88(H)x94.12(V)x7.7(D)	mm	Note1
11	PanelActive Area 可视区域	89.76X83.49	mm	Note1
12	PixelPitch 像素尺寸	0.055x0.165	mm	-
13	PixelArrangement 像素排列	RGB-stripe		-
14	Weight 重量	TBD	g	-
15	Driver IC 驱动芯片	ST72566	-	-
17	LightSource 背光源	20 LED light	-	-
18	Interface 接口方式	RGB 24BIT	-	-
19	Operating Temperature 工作温度	-30~+85	℃	-
20	Storage Temperature 存储温度	-30~+85	℃	-

Note 1: Please refer to the mechanical drawing

注1：请参照模组图；

2.Pin Assignments 接口定义

Pin No. Pin 序号	Symbol符号	Function 功能描述
1	LED K	Backlight cathode 背光负极输入端。
2	LED A	Backlight anode 背光正极输入端
3	GND	Ground 接地
4	VCC	Analog Power Supply for LCM （2.8V） 系统输入电压2.8V。
5-12	R0-R7	Data bus
13-20	G0-G7	Data bus
21-28	B0-B7	Data bus
29	GND	Ground 接地
30	CLK	Pixel clock signal in RGB I/F
31	DISP	Display
32	HSYNC	Horizontal sync .Signal in RGB I/F
33	VSYNC	Vertical sync .Signal in RGB I/F
34	DEN	Data enable signal in RGB I/F DE mode
35	NC	NC
36	GND	Ground 接地
37	NC (XR)	NC
38	NC (YD)	NC
39	NC (XL)	NC
40	NC (YU)	NC

3. Electrical Specification 电气特性

3.1 Absolute Maximum Ratings 极限参数

Item	Symbol	Min.	Typ.	Max.	Unit	Conditions
Supply Voltage	VDD	3.0	3.3	3.6	V	
IO Supply Voltage	VDDI	3.0	3.3	3.6	V	
Charge Pump Supply Voltage	PVDD	3.0	3.3	3.6	V	

3.2 Typical Operation Conditions 典型工作条件

Item	Symbol	Min.	Typ.	Max.	Unit	Conditions
Logic-High Input Voltage	Vih	0.7VDDI	-	VDDI	V	
Logic-Low Input Voltage	Vil	DGND	-	0.3VDDI	V	
Logic-High Output Voltage	Voh	VDDI-0.4	-	VDDI	V	
Logic-Low Output Voltage	Vol	DGND	-	DGND+0.4	V	

3.3 DC Characteristics for Analog Circuit

Item	Symbol	Min.	Typ.	Max.	Unit	Conditions
Positive High-Voltage Power	VGH_S	8	15	17	V	No Load@ FR=60Hz
Negative High-Voltage Power	VGL	-12	-10	-7	V	
Output Voltage Deviation	Vod		±40	±50	mV	
Standby Current	Isc	-	-	50	uA	
Operation Current	Ioc	-	50	-	mA	

3.4 Backlight Circuit Characteristics 背光功耗

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current 背光电流	I_B		180	-	mA
LED Voltage 背光电压	Vf		15		V
Power Consumption 功耗	P_{BL}	-		-	mW

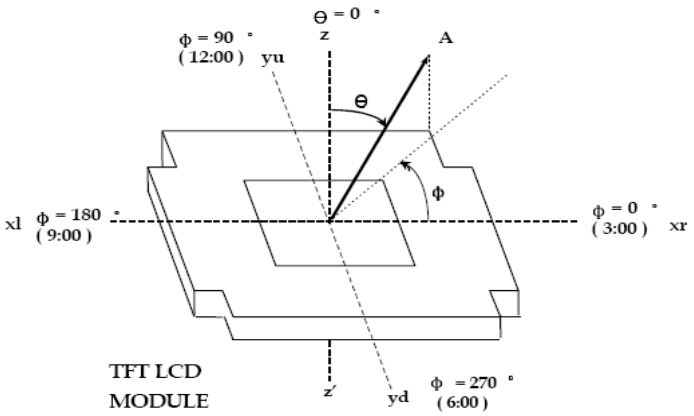
4.Optical Specification 光学参数

4.1LCM Optical Characteristics 液晶模组光学特性

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Viewing Angle Range 视角	Left	θ_L	CR≧10	80	85	-	degree
	Right	θ_R		80	85	-	
	Top	θ_T		80	85	-	
	Bottom	θ_B		80	85	-	
Response Time 响应时间		Ton+Toff	θ=Φ=0°	-		40	ms
Contrast Ratio 对比度		CR	θ=Φ=0°	1000	1200	-	-
Luminance 亮度		L	θ=Φ=0°	-	1000	-	cd/m^2
Color Chromaticity (CIE1931) 色坐标	White	W_x	Normal θ=Φ=0°		0.293		-
		W_y			0.336		
	Red	R_x			0.653		
		R_y			0.323		
	Green	G_x			0.281		
		G_y			0.563		
	Blue	B_x			0.136		
		B_y			0.143		
Uniformity 均匀度		U_L	θ=Φ=0°	-	80	-	%
Flicker 闪烁		-	-	No Visible			-

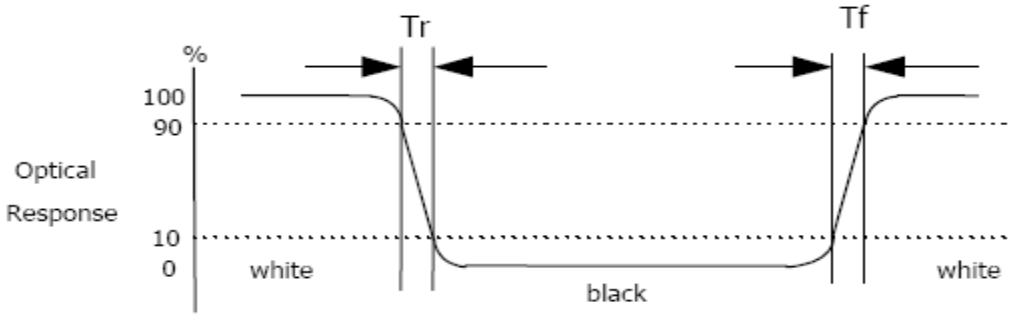
4.2 Measurement system 测量系统

4.2.1 LCM Viewing Angle



Viewing angle is the angle at which the contrast ratio is greater than 10.The angles are determined f horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surfa

4.2.2 Response time



Response time is the time required for the display to transition from white to black (Rising time, Tr) and to white (Falling time, Tf) for additional information.

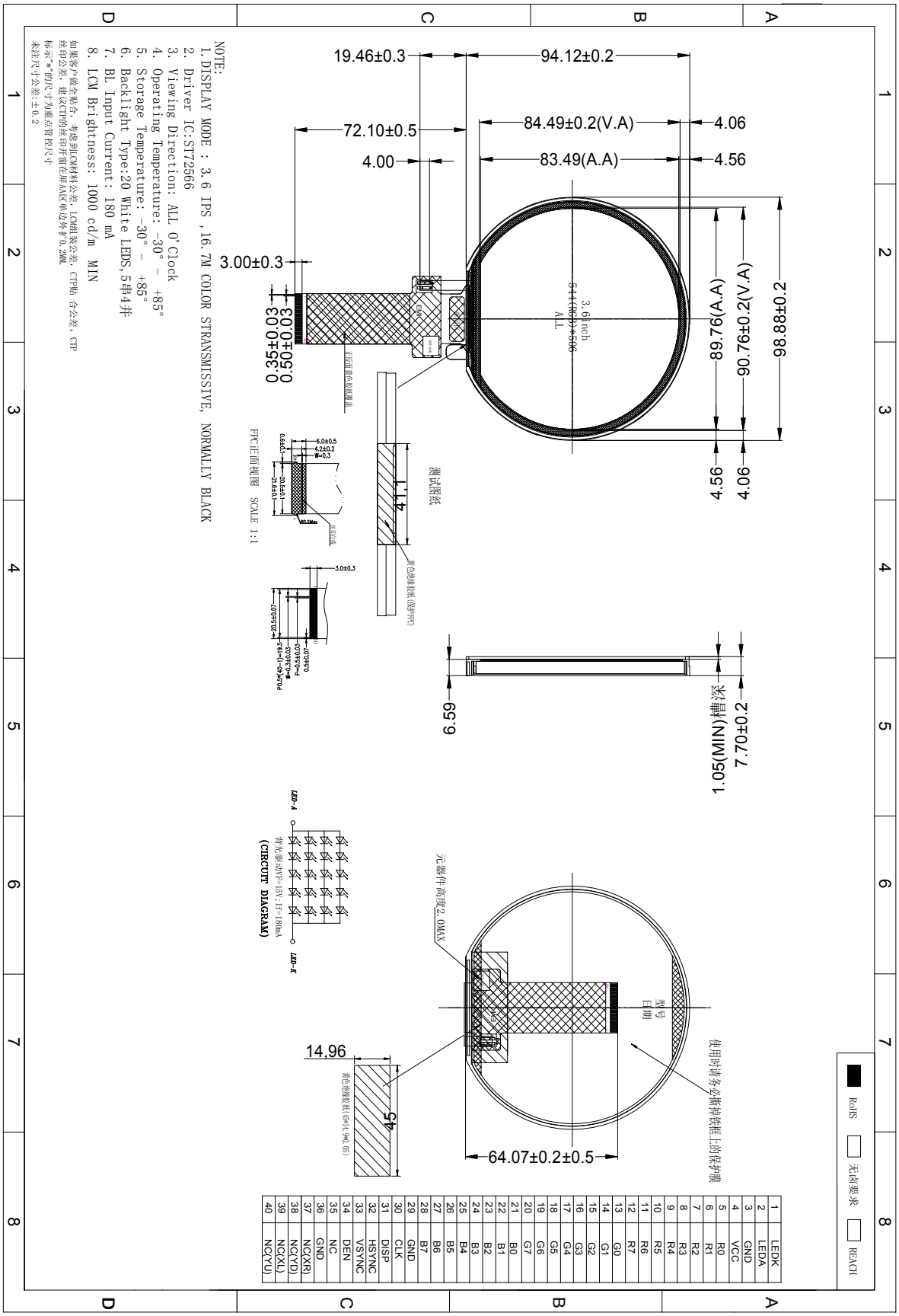
4.2.3 Contrast Ratio (CR)

Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

Surface luminance is the center point across the LCD surface 500mm from the surface with all pixels displayin white.

5. Mechanical Drawing 模组图

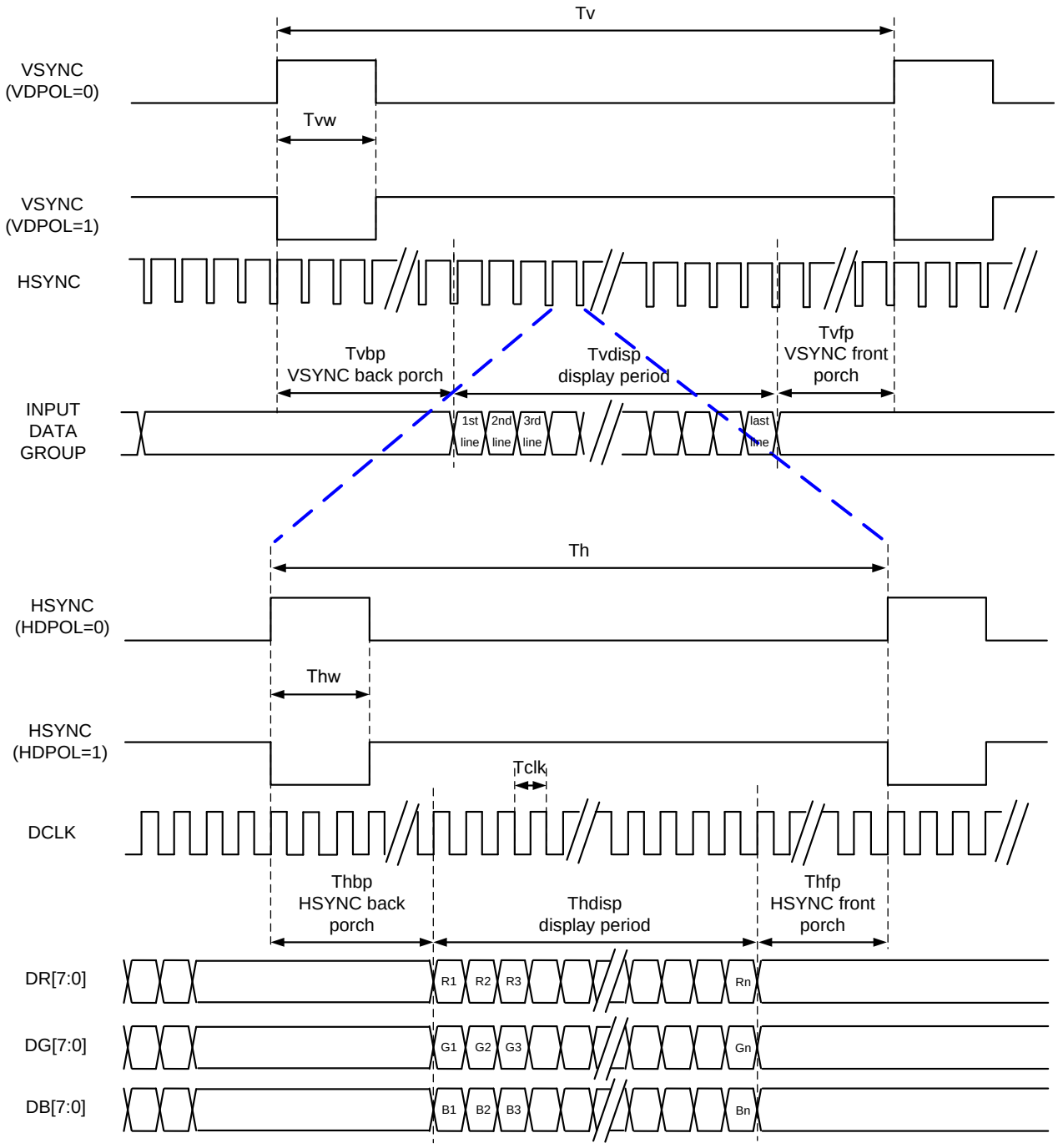


6. RGB Interface RGB 模式

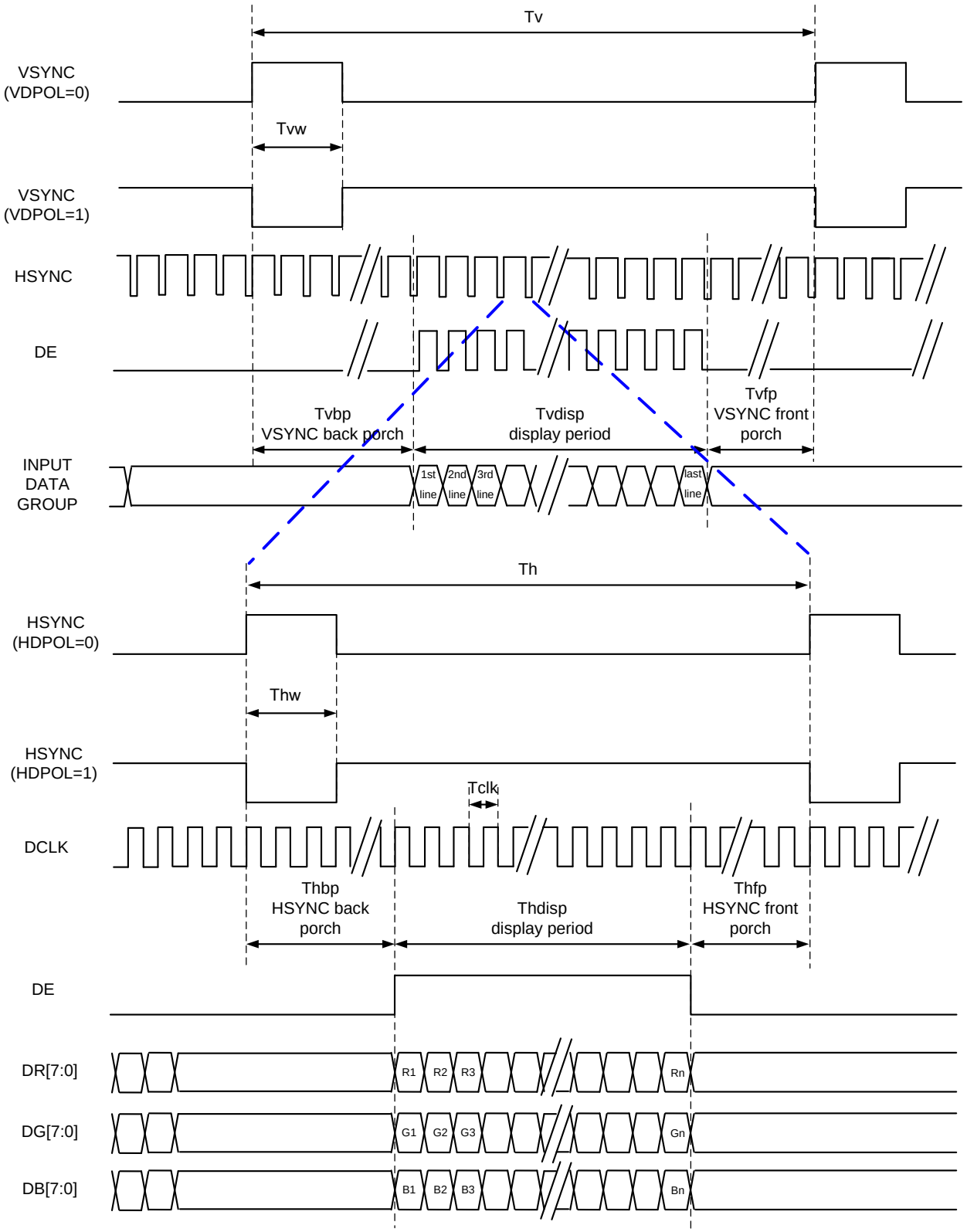
RGB Mode Selection Table	DCLK	HSYNC	VSYNC	DE
SYNC - DE Mode	Input	Input	Input	Input
SYNC Mode	Input	Input	Input	GND
DE Mode	Input	Input	GND	Input

Note: "Input" means these signals are driven by host side

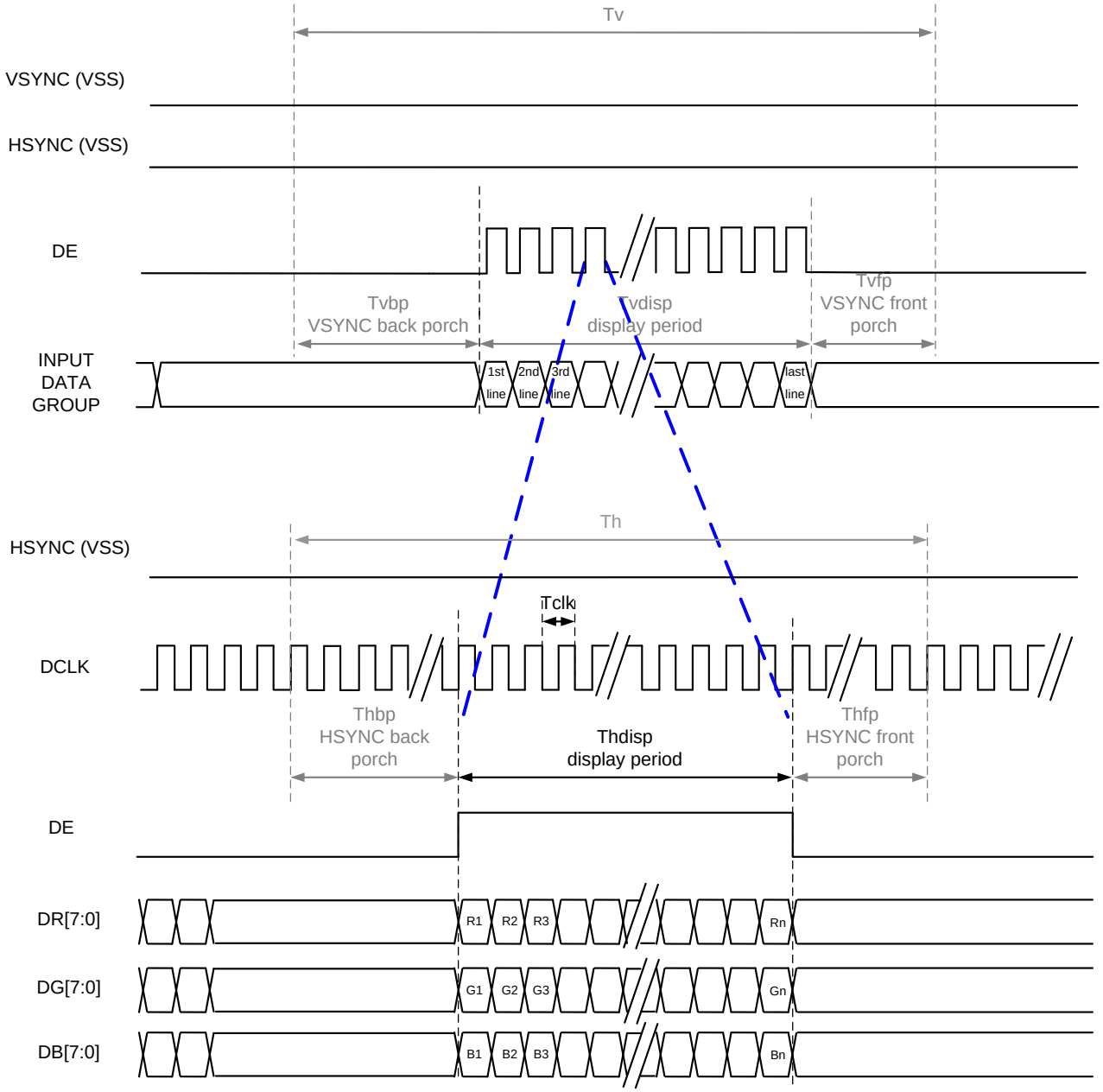
6.1 SYNC Mode



6.2 SYNC-DE Mode

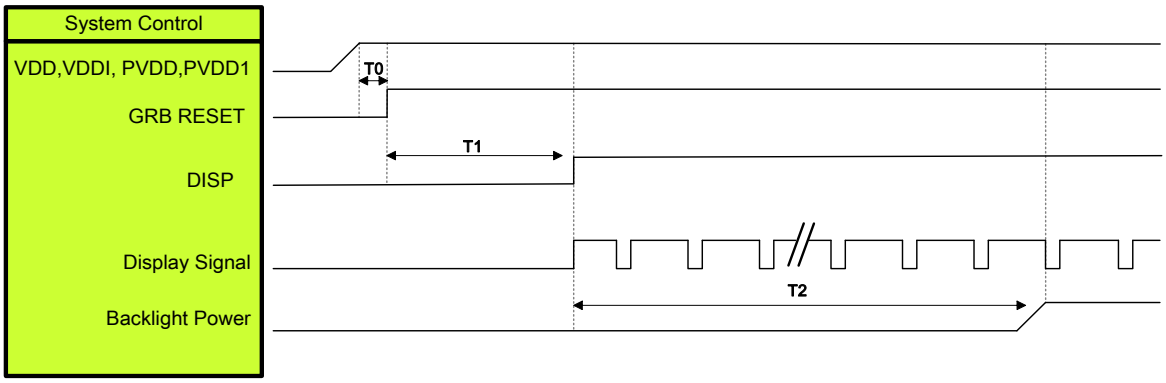


6.3 DE Mode


Tv
VSYNC (VSS)
HSYNC (VSS)
DE
Tvbp
VSYNC back porch
Tvdisp
display period
Tvfp
VSYNC front
porch
INPUT
DATA
GROUP
1st
line
2nd
line
3rd
line
last
line
Th
HSYNC (VSS)
Tclk
DCLK
Thbp
HSYNC back
porch
Thdisp
display period
Thfp
HSYNC front
porch
DE
DR[7:0]
R1
R2
R3
Rn
DG[7:0]
G1
G2
G3
Gn
DB[7:0]
B1
B2
B3
Bn

7. POWER ON/OFF SEQUENCE 上电时序

7.1 Power On Sequence



Symbol	Description	Min. Time	Unit
T0	System power stability to GRB RESET signal	0	ms
T1	GRB RESET= “High” to DISP=”High”	10	ms
T2	Display Signal output to Backlight Power on	250	ms

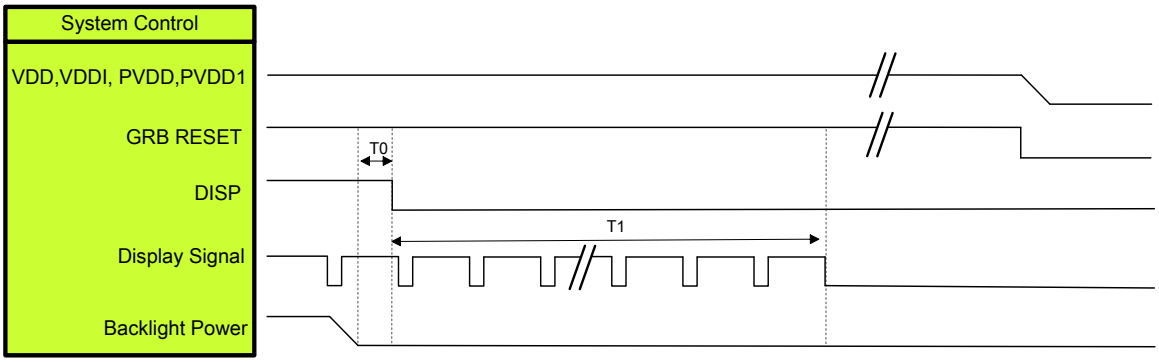
Note :

1. When DISP pull “H” or “L”, IC will execute the internal power on or power off procedures .Please be careful about the timing of DISP and do not interrupt it during power on or power off procedure, otherwise unexpected errors will occur.

2. RGB interface Display signal: DCLK; VSYNC; HSYNC; DE; DR[7:0]; DG[7:0]; DB[7:0]

3: LVDS interface Display signal: DCLK P/N; RX[3:0] P/N

7.2 Power Off Sequence



Symbol	Description	Min. Time	Unit
T0	Backlight Power off to DISP=”Low”	5	ms
T1	DISP=”Low” to IC internal voltage discharge complete	100	ms

Note :

1. When DISP pull “H” or “L”, IC will execute the internal power on or power off procedures .Please be careful about the timing of DISP and do not interrupt it during power on or power off procedure, otherwise unexpected errors will occur.

2. RGB interface Display signal: DCLK; VSYNC; HSYNC; DE; DR[7:0]; DG[7:0]; DB[7:0]

3: LVDS interface Display signal: DCLK P/N; RX[3:0] P/N

8. Reliability Test Items 可靠性测试项目

Test Item 测试项目	Test Condition 测试条件	Test result determinant gist 实验结果判定
High temperature storage 高温存储	85±3℃，24H ；	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 试验结束后,已测试的LCD 样品必须在室内正常温湿度环境下放置2~4个小时以上才能进行功能和外观检查，样品不允许有以下缺陷： 1.Air bubble in the LCD; 模块中有气泡； 2.Non-display;不显示； 3.Glass crack;玻璃破碎； 4. The electrical characterist requirements shall be satisfie 需要满足模块电气性能。
Low temperature storage 低温存储	-35±3℃，24H ；	
High temperature operation 高温运行测试	85±3℃，24H ；	
Low temperature operation 低温运行测试	-35±3℃，24H ；	
High temperature /humidity高温高湿	60℃±3℃,90%±3%RH ，24H ；	
Thermal Shock 冷热冲击	-35℃/0.5h~+85℃/0.5h for a total24 cycles;	
Vibration Test 振动测试	Frequency 10Hz~55Hz~10Hz Amplitude ：1.5mm，X，Y，Z direction for total; 1H (Packing condition)	
ESD test 静电测试	±2KV, Human Body Mode, 150pF/330Ω ；±8KV, Air Mode, 150pF/330Ω	

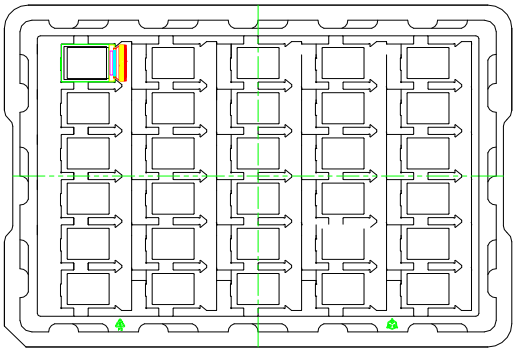
Remark: 注意：

1. The test samples should be applied to only one test item.
   每个被测试的模块只能用于其中的一个测试项目。
2. Sample size for each test item is 2pcs.
   每个测试项目的样品数量为2片。
3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.
   故障判断标准:基本规格,电气特性,机械特性,光电特性。

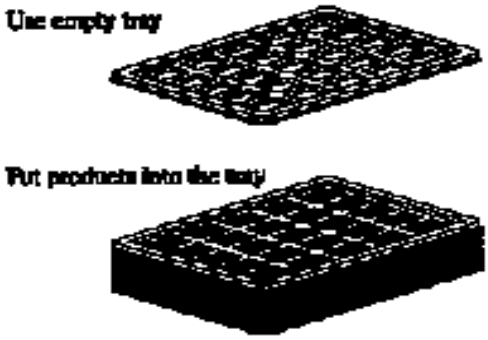
9. Packing and Storage Specification(Reference Only)包装存储

9.1 Packing Method 包装方法

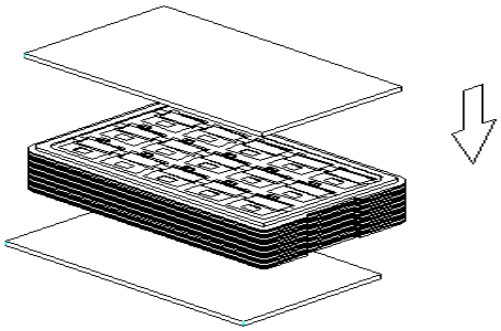
(1)



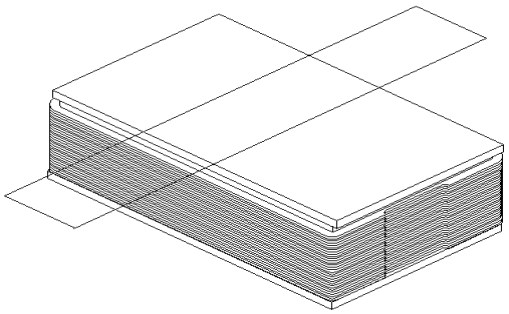
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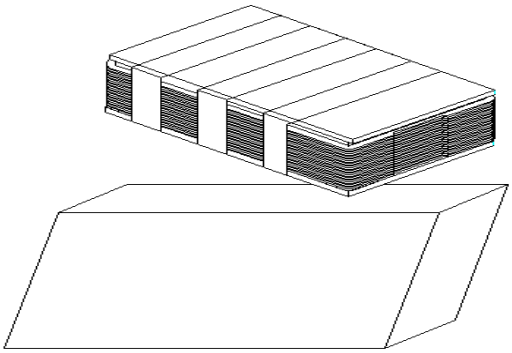
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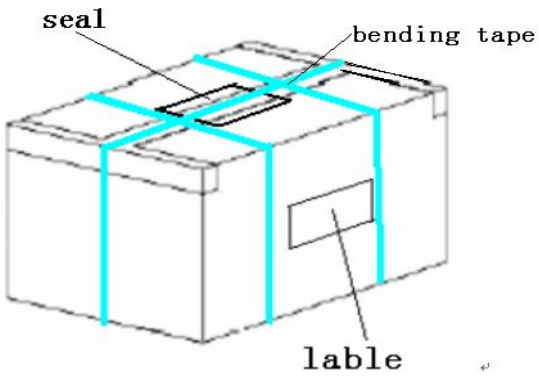
(4)



(5)



(6)



1. Put module into tray cavity. 把模块放进托盘
2. Tray stacking. 托盘叠装
3. Put 1 foam under the tray stack and 1 foam above. 在托盘上下放卡板
4. Fix the cardboard to the tray stack with adhesive tape. 绑胶带
5. Put the tray stack into carton. 把邦好的托盘放进纸箱
6. Carton sealing with adhesive tape. 封纸箱

9.2 Storage Method 存储方法

1. Store in an ambient temperature of 23 ℃±5 ℃, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.
   存储环境温度为 23±5 ℃，相对湿度为 55%±15% ，存储不能超过 12 个月，不要长时间暴晒。
2. Store in a clean environment, free from dust, active gas, and solvent.
   存储在一个干净的环境，不受灰尘，活性气体和溶剂污染。
3. Store in antistatic container.
   存储在防静电环境。

10. Announcements 注意事项

1. Do not attempt to disassemble or process the LCD module.
   请勿拆卸液晶显示模块。
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached.
   不要在印制电路板上钻额外的孔，修改形状或更改印制线路板上元件的位置。
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron welding temperature at 320 ° C to 350 ° C, the welding timecontrol within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
   除焊接接口外，不要用烙铁做任何更改；焊接温度保证在 320 ℃-350 ℃，焊接时间控制在 10S 以内，焊接时注意不要在同一处停留时间太久以免烫伤 FPC 。
4. Other matters in not clear before use, please contact our staff to guide.
   其他事项在不清楚使用之前，请联系我司人员指导进行。